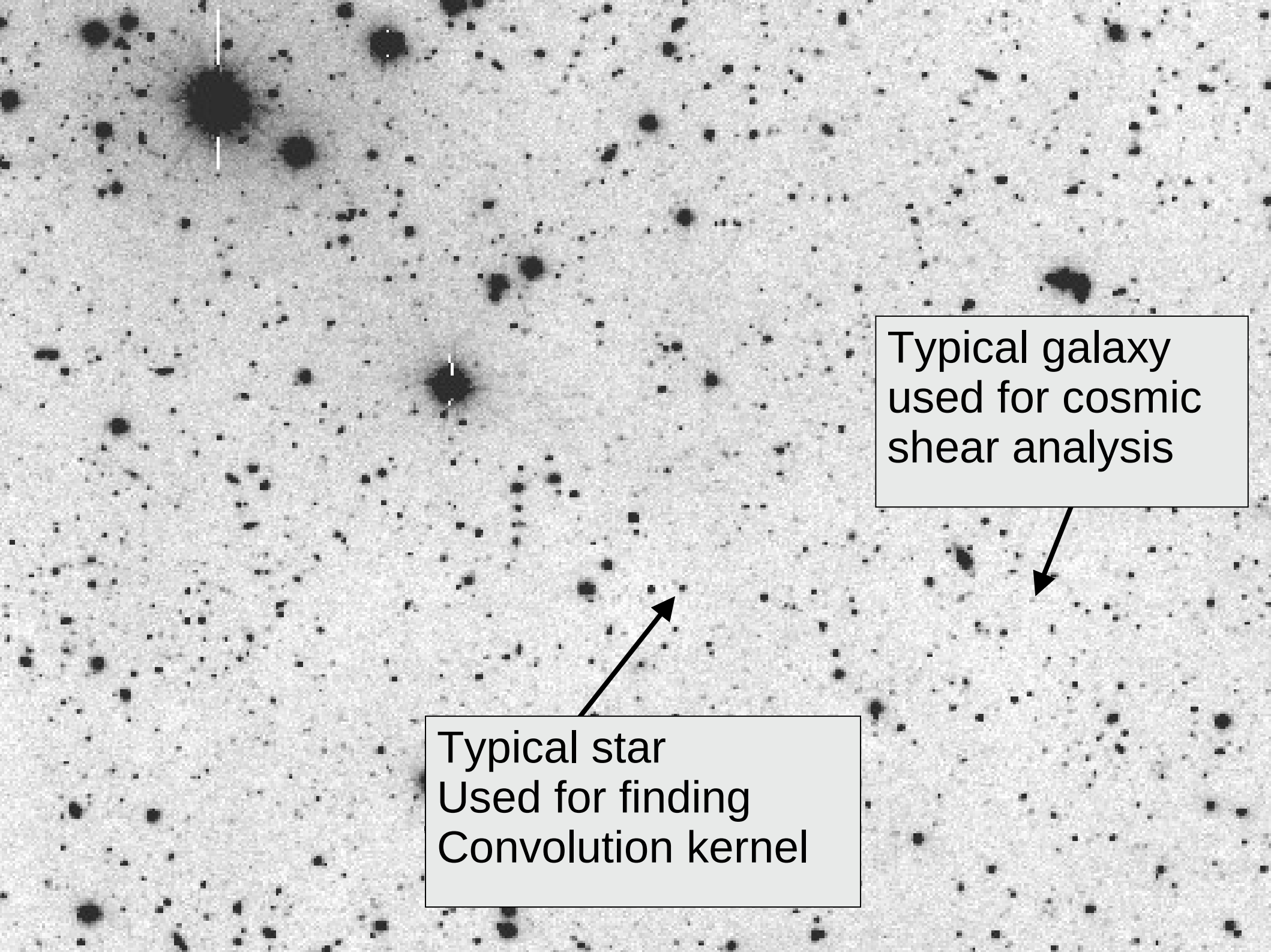




Typical galaxy
used for cosmic
shear analysis

Typical star
Used for finding
Convolution kernel

Cosmic Lensing



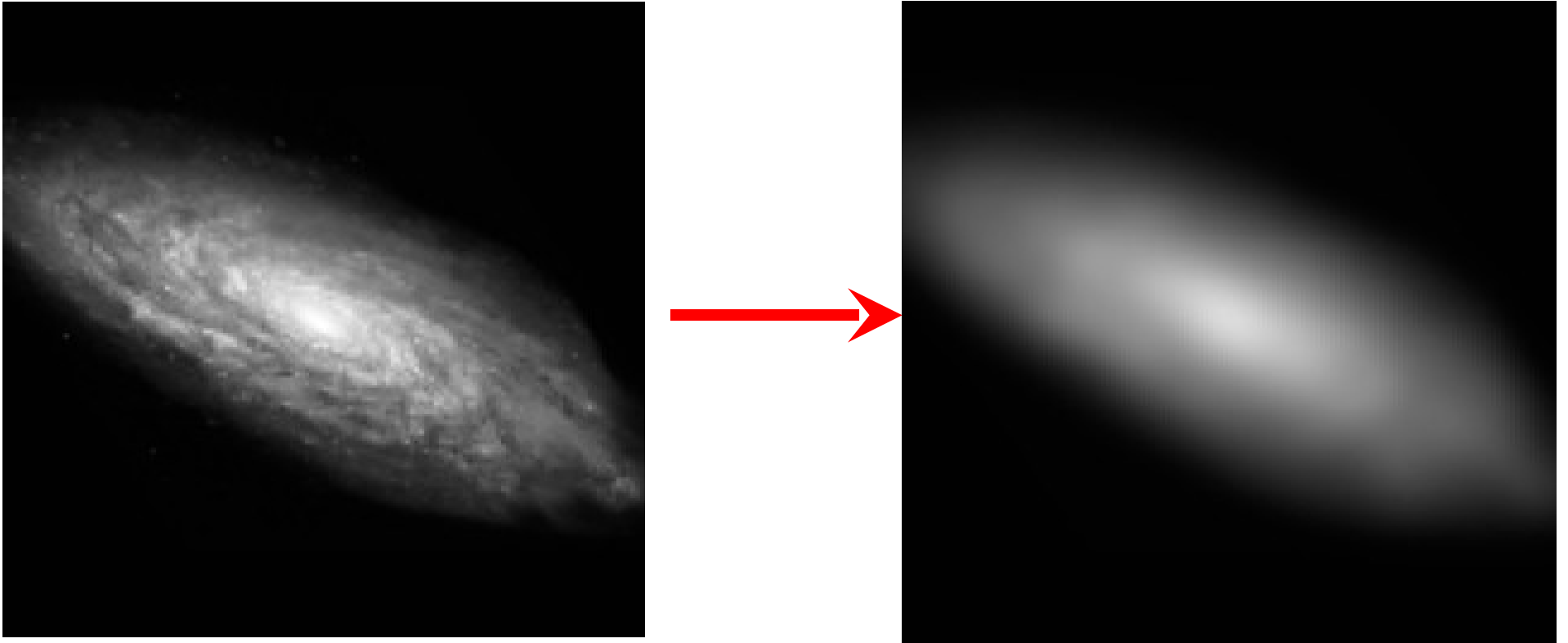
$g_i \sim 0.2$



$$\begin{pmatrix} x_u \\ y_u \end{pmatrix} = \begin{pmatrix} 1 - g_1 & -g_2 \\ -g_2 & 1 + g_1 \end{pmatrix} \begin{pmatrix} x_l \\ y_l \end{pmatrix}$$

Real data:
 $g_i \sim 0.03$

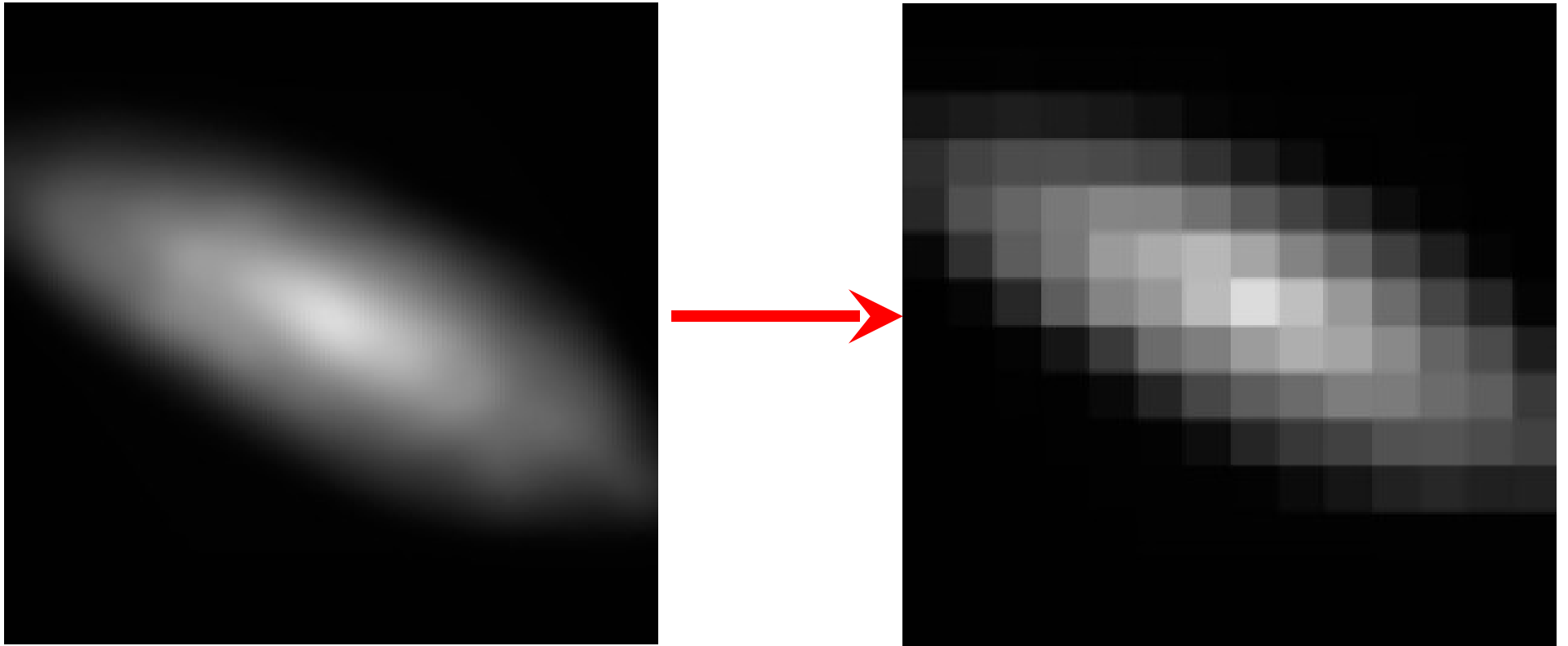
Atmosphere and Telescope



Convolution with kernel

Real data: Kernel size $\sim$ Galaxy size

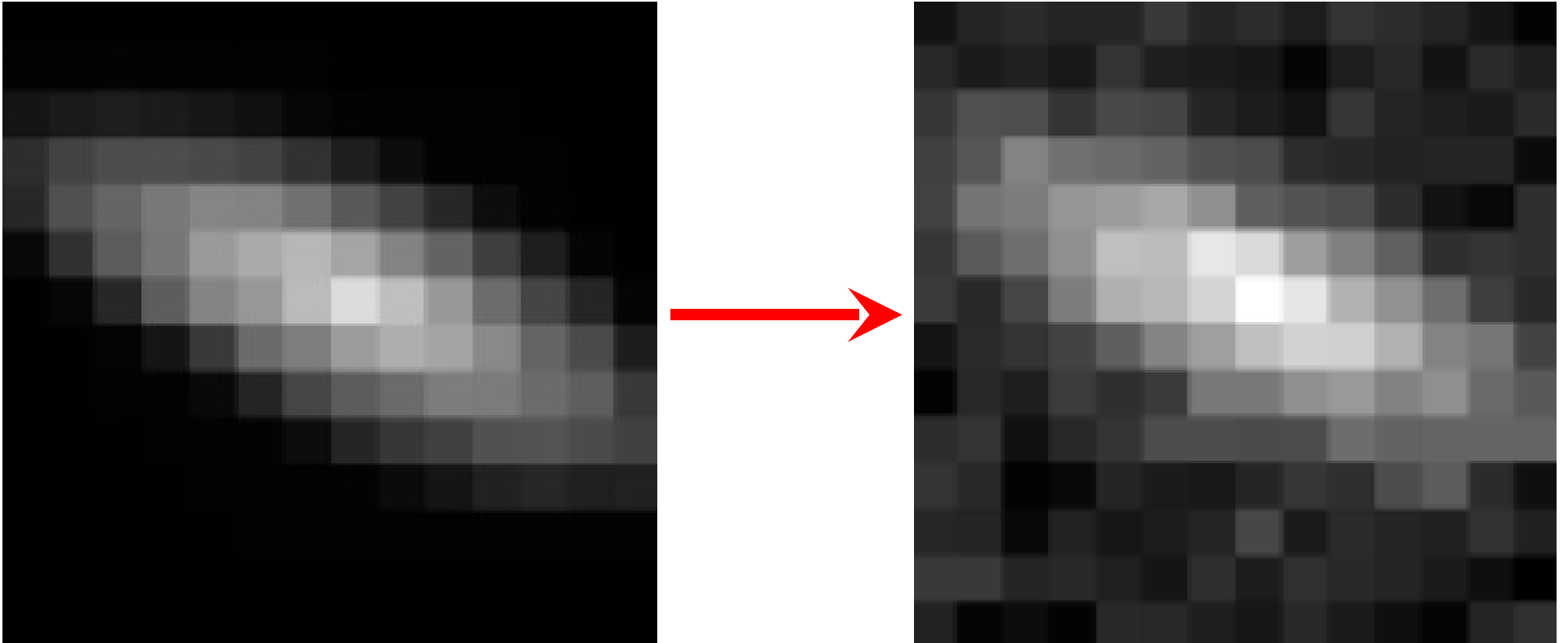
Pixelisation



Sum light in each square

Real data: Pixel size $\sim$ Kernel size / 2

Noise



Mostly Poisson. Some Gaussian and bad pixels.
Uncertainty on total light ~ 5 per cent

